

6(a)(i)	(electron) neutrino	B1
6(a)(ii)	weak (nuclear force/interaction)	B1
6(a)(iii)	β^+ (particle)	B1
6(a)(iv)	(quark structure is) up up down or uud	B1
	$(2/3)e + (2/3)e - (1/3)e = (+)e$	B1
6(a)(v)	up up down changes to up down down or uud $\rightarrow$ udd or up changes to down or u $\rightarrow$ d	B1
6(b)(i)	$F = Eq$	C1
	ratio = $6/8$ = 0.75	A1
6(b)(ii)	ratio = $0.75 \times (16/12)$ = 1.0	A1
6(b)(iii)	horizontal straight line at a non-zero value of a	B1